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Local Plug-and-Play Compatible Stability Conditions for AC Grids

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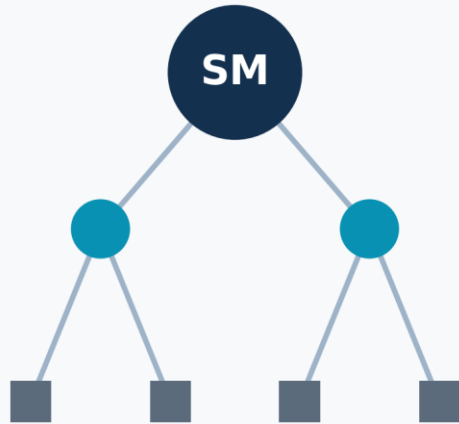
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The grid is changing faster than our analysis tools

Few large machines, top-down flow



Many small inverters, meshed & changing



1

Inertia is disappearing

Synchronous generation → grid-forming power electronics.

2

New devices and control, faster timescales

Complex inverter control schemes replace electromechanical dynamics.

3

Topology in constant flux

From top-down to highly distributed; participants and topology change frequently.

4

Monolithic analysis breaks

Eigenvalues of a fixed, system-wide model are no longer appropriate.

Needed: Stability guarantees that do not require re-analysing the whole grid any time the network interconnection changes

Why local conditions?

Goal: certify grid-wide small-signal stability via frequency-domain quadratic constraints on the input–output properties of local grid subsystems.

BENEFITS

- Simplifies analysis — from top-down to bottom-up
- Basis for interconnection standards & grid codes for inverter-dominated grids
- Frequency-domain $\Rightarrow$ works with black-box models; no proprietary disclosure
- **Plug-and-play: a compliant device connects without re-running global analysis**

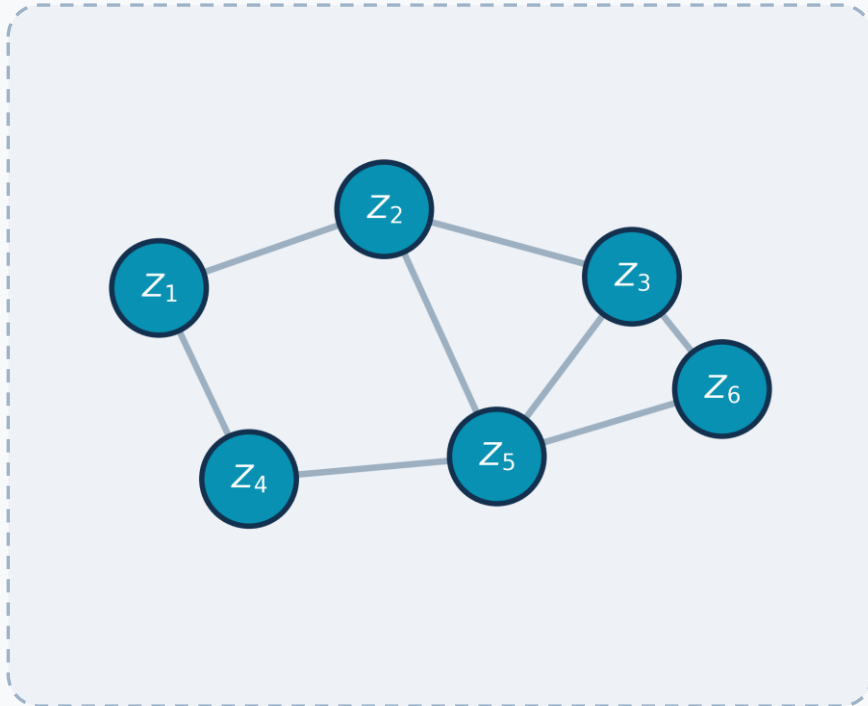


GAPS IN THE LITERATURE

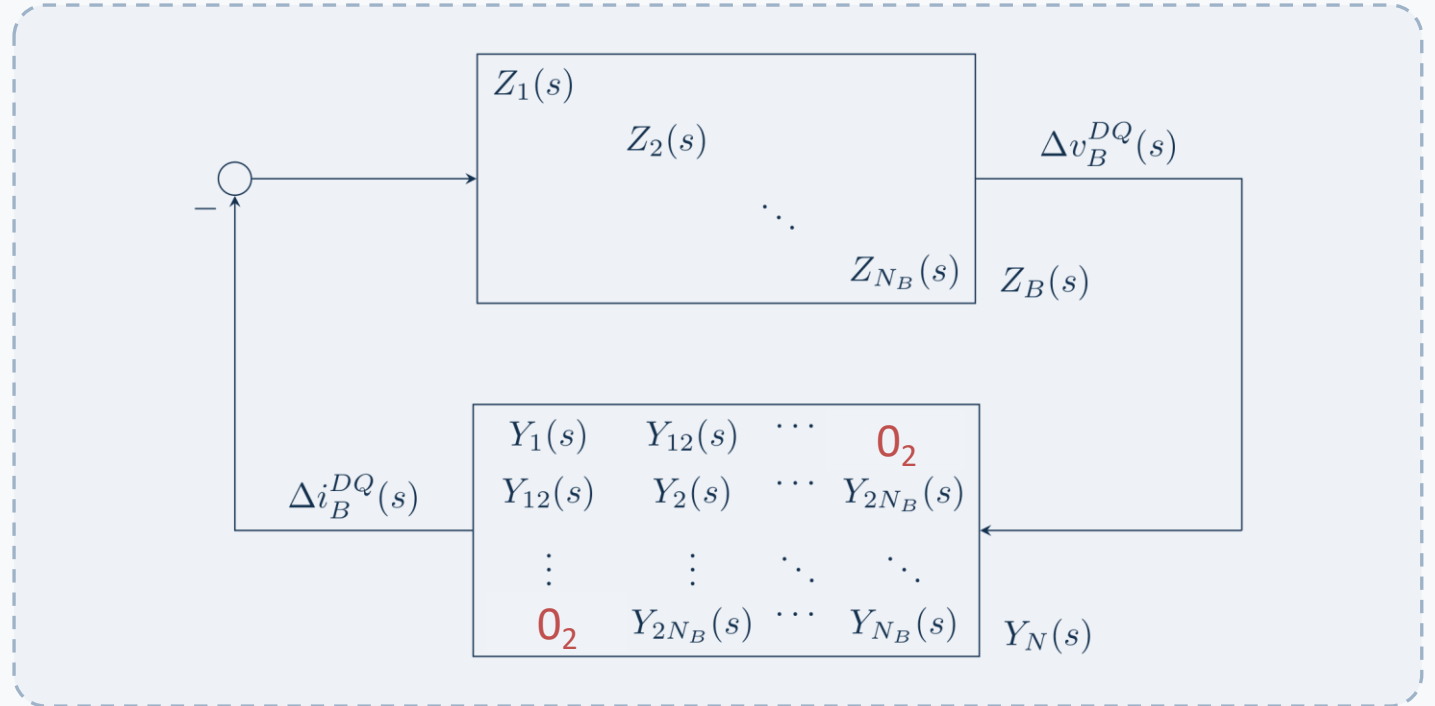
- Simplified bus/network models ignore real complexity
- Centralised knowledge of topology required $\rightarrow$ forfeits plug-and-play

Huang et al., "Gain and Phase," IEEE Trans. Power Syst., 2024 · Häberle et al., "Decentralized Parametric Stability Certificates," arXiv, 2025

Grid model: block-diagonal buses, sparse network



Graph: buses (nodes), power lines (edges); signals in a common DQ frame.



Feedback loop of buses (block-diagonal) and network (sparse).

Kirchhoff's laws → stability of the feedback pair

$$\begin{bmatrix} Z_B(s), & Y_N(s) \end{bmatrix}$$

From the Nyquist criterion to quadratic constraints

Stable buses and network: grid stable $\Leftrightarrow$
characteristic loci of the return ratio do not
encircle -1

$$L(s) = Y_N(s) Z_B(s)$$

SUFFICIENT CONDITION — HOMOTOPY

$$-1 \notin \sigma(\tau L(j\omega)) \quad \forall \tau \in [0, 1], \quad \forall \omega \geq 0$$

Holds at ω if there exist multipliers $\Pi(j\omega)$ — free parameters, re-chosen at each frequency — satisfying:

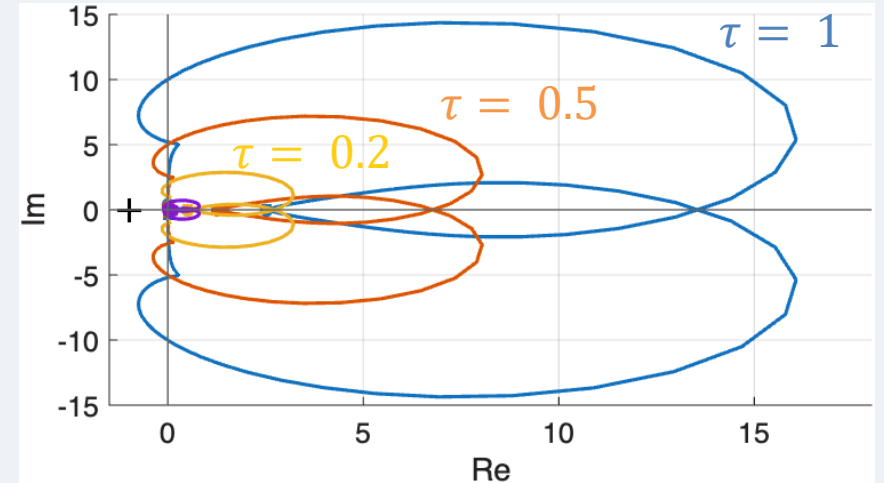
NETWORK CONSTRAINT — ADMITTANCE

$$\begin{bmatrix} I \\ -Y_N(j\omega) \end{bmatrix}^* \begin{bmatrix} \Pi_{11} & \Pi_{12}^* \\ \Pi_{12} & \Pi_{22} \end{bmatrix} \begin{bmatrix} I \\ -Y_N(j\omega) \end{bmatrix} \leq 0$$

$$\Pi_{11} = \Pi_{11}^* \leq 0, \quad \Pi_{22} = \Pi_{22}^* \geq 0$$

BUS CONSTRAINT — IMPEDANCES

$$\begin{bmatrix} Z_B(j\omega) \\ I \end{bmatrix}^* \begin{bmatrix} \Pi_{11} & \Pi_{12}^* \\ \Pi_{12} & \Pi_{22} \end{bmatrix} \begin{bmatrix} Z_B(j\omega) \\ I \end{bmatrix} > 0$$



Loci shrink to 0 without touching $-1 \Rightarrow$ no encirclements.

Decentralised conditions — one certificate per bus

Exploit the block-diagonal bus structure: find block-diagonal multipliers, one 2×2 block per bus $\Rightarrow$ the bus constraint decouples:

$$\begin{bmatrix} Z_i(j\omega) \\ I_2 \end{bmatrix}^* \begin{bmatrix} \Pi_{11}^i & \Pi_{12}^{i*} \\ \Pi_{12}^i & \Pi_{22}^i \end{bmatrix} \begin{bmatrix} Z_i(j\omega) \\ I_2 \end{bmatrix} > 0 \quad \forall \nu_i \in \mathcal{V}_B$$

Multiplier blocks independent of non-local parameters $\rightarrow$ fully decentralised constraint.

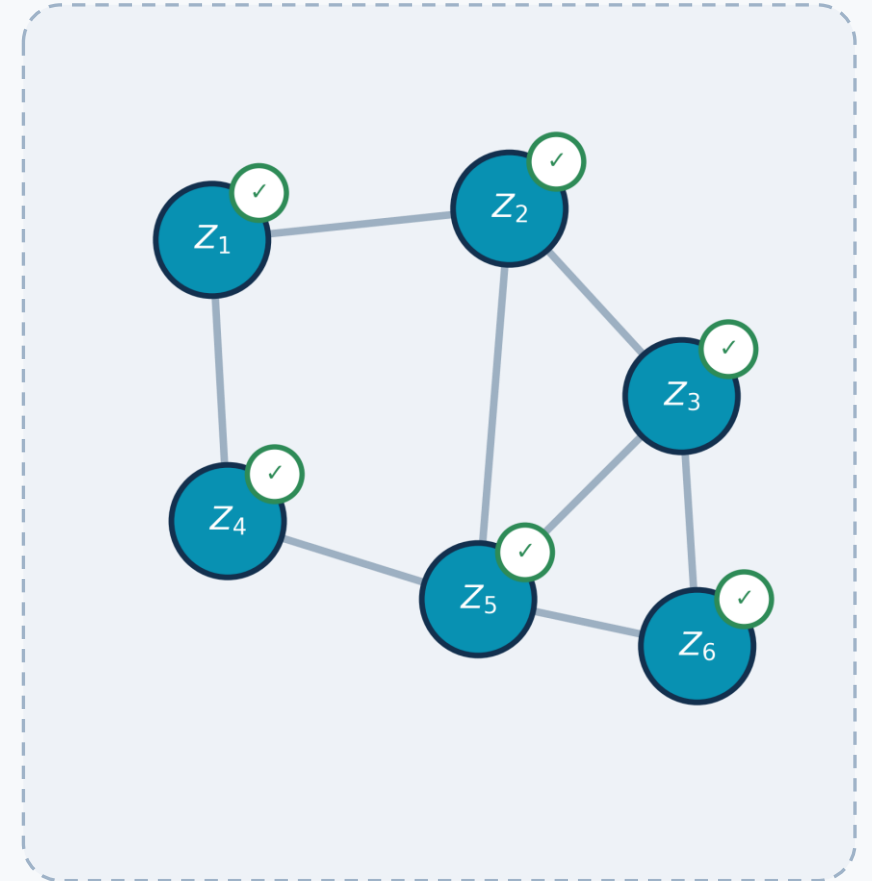
SIMPLE EXAMPLE — POSITIVE-REAL

Network is passive $\rightarrow$ Positive-real $\rightarrow$ Network constraint satisfied for all ω

Require:

$$\Pi_{11}^i = \Pi_{22}^i = 0_2, \quad \Pi_{12}^i = I_2 \implies Z_i(j\omega) + Z_i(j\omega)^* > 0$$

Uses no network-level information $\rightarrow$ plug-and-play.



Decentralised extended subsystem conditions

Exploit network sparsity:

- Regroup into extended subsystems
- Each bus together with all its connected lines
- In feedback with a static interconnection:

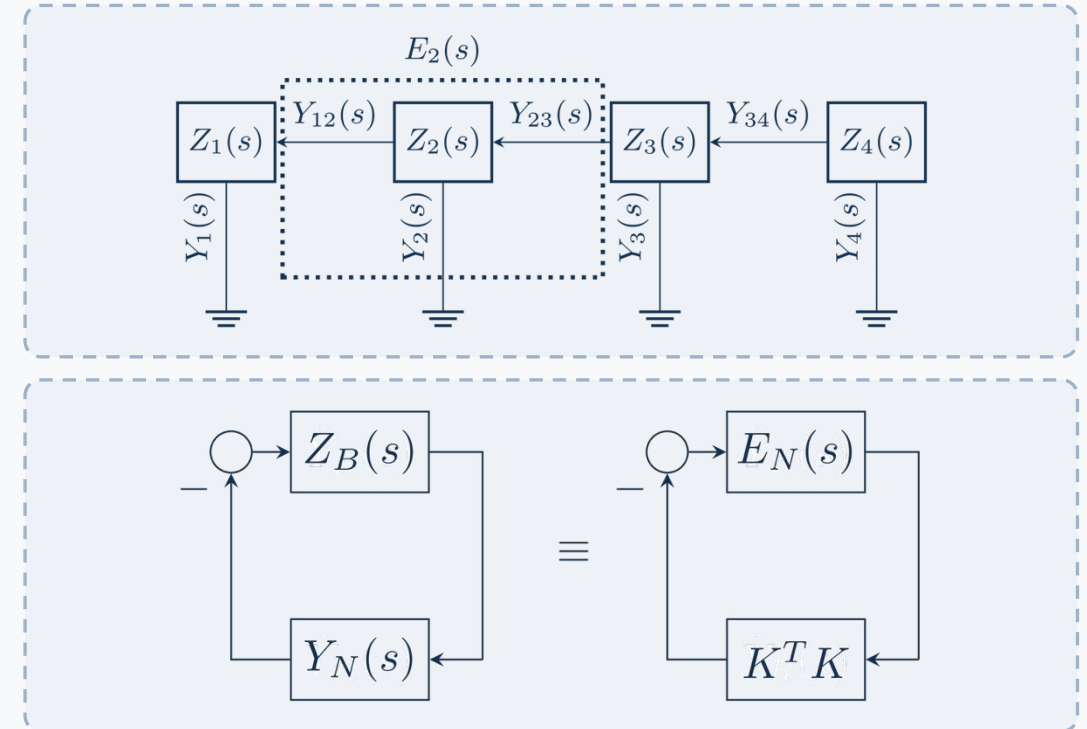
$$\sigma(Y_N(s) Z_B(s)) \sim \sigma(K E_N(s) K^T)$$

$$E_N(s) = \text{diag}(E_1(s), \dots, E_{N_B}(s))$$

- Homotopy on this loop $\Rightarrow$ homotopy on the original loop.
- The resulting local conditions:

At each line $(v_i, v_j) \in \mathcal{E}_{\mathcal{P}}$:

$$-\Pi_{12}^{ij} - \Pi_{12}^{ij*} + 2\Pi_{22}^{ij} \leq 0$$



On each extended subsystem:

$$\begin{bmatrix} E_i(j\omega) \\ I \end{bmatrix}^* \begin{bmatrix} 0 & \hat{\Pi}_{12}^{i*} \\ \hat{\Pi}_{12}^i & \hat{\Pi}_{22}^i \end{bmatrix} \begin{bmatrix} E_i(j\omega) \\ I \end{bmatrix} > 0 \quad \forall v_i \in \mathcal{V}_B$$

Network changes only affect nearby constraints $\rightarrow$ plug-and-play preserved

Main result, plug-and-play, and a case study

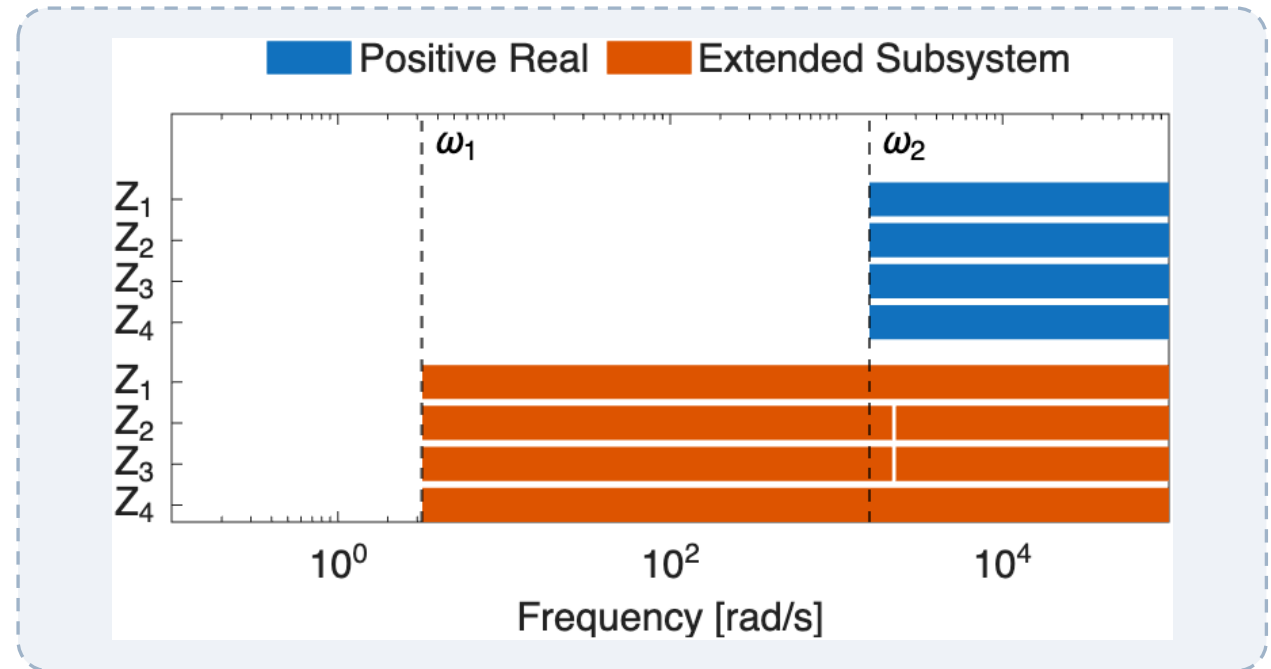
Proposition: The grid is stable if, at every $\omega \geq 0$, either

- the decentralised bus conditions; OR
- the extended subsystem conditions hold, and the choice may differ across frequency ranges.

FROM THEOREM TO GRID CODE

- Interconnection standard: conditions any new bus must meet before connecting
- Centralised analysis usable when the decentralised framework is infeasible

CASE STUDY — KUNDUR TWO-AREA SYSTEM



- Plug-and-play conditions certify $\omega > 10^0$ rad/s — below the problematic range
- Lower ω : more centralised analysis (recent work: typically trivially satisfied via power flow)

Conclusions & future work



Local certification, global guarantee

Quadratic constraints on local subsystems collectively enforce the generalised Nyquist criterion.



Flexible, less conservative

Unsimplified dynamics; conditions vary across frequency ranges and system formulations.



A foundation for plug-and-play grid codes

Compliant devices connect without global re-analysis.

Recent extensions:

- accounting for unstable bus dynamics
- alternative representations capturing power-flow behaviour at low frequencies
- graphical interpretations

Thank you — questions welcome. lh706@cam.ac.uk